

Optimizing Direct Marketing

Abstract

The Banking Industry heavily relies on marketing to maximise profits through increased customer engagement and outreach. In this study, we aim to address the critical challenge of optimizing direct marketing campaigns for banking institutions by predicting customer subscription to a term deposit. Effective targeting is essential for minimizing resource expenditure and maximizing campaign success. Using a real-time Bank Marketing dataset, we developed a predictive model to examine and identify factors that influence a customer's decision to subscribe, thus enabling targeted outreach. The research is guided by the central question of whether indicators of financial stability and customer engagement meaningfully shape subscription behavior.

Drawing from the dataset, we applied extensive preprocessing, feature engineering, and a series of classification models to evaluate the predictive importance of these factors. Special attention was given to managing class imbalance and constructing variables that capture customer financial capacity and campaign interaction patterns. The models were tuned and compared to ensure robust assessment of the hypothesized relationships.

The findings provide empirical support for the hypothesis: clients with higher balances, no personal loans, longer last-contact durations, and successful previous campaign outcomes show a higher likelihood of term deposit subscription. These insights reveal how financial stability and engagement jointly influence campaign success and highlight the variables most useful for improving targeting strategies.

Overall, the study demonstrates how predictive modeling can translate behavioral and financial indicators into actionable guidance for more precise and effective marketing campaigns in the banking sector.

Introduction

Direct marketing has become an increasingly important component of modern banking operations, particularly as financial institutions seek to improve customer acquisition and strengthen long-term product adoption. In an environment where consumers are continuously exposed to competing financial offers, banks face the challenge of identifying which clients are most receptive to term deposit products. The ability to distinguish between likely and unlikely subscribers is not only a matter of marketing efficiency but also a strategic necessity for reducing operational costs and improving customer experience.

Research in bank marketing analytics highlights several factors that may shape subscription behavior, including indicators of financial stability, demographic characteristics, and engagement during prior campaigns (Moro, Cortez & Rita, 2014; Moro et al., 2011). Existing studies emphasize that variables such as account balance, loan status, call duration, and outcomes of previous campaigns can meaningfully influence customer decisions. However, many earlier investigations report methodological limitations, including challenges with class imbalance, inconsistent preprocessing, and limited attention to how financial and behavioral variables interact. These gaps suggest the need for a more structured analysis that examines these predictors collectively rather than in isolation.

This study contributes to that gap by evaluating whether customers with higher financial capacity and stronger engagement indicators are more likely to subscribe to a term deposit. While the detailed modelling results are presented in later sections, the introduction of predictive analytics allows for a more nuanced characterization of subscription tendencies. By situating the analysis within established literature and addressing recurring methodological issues, this work aims to provide a clearer understanding of the customer traits most relevant for improving direct marketing strategies in the banking sector.

Result

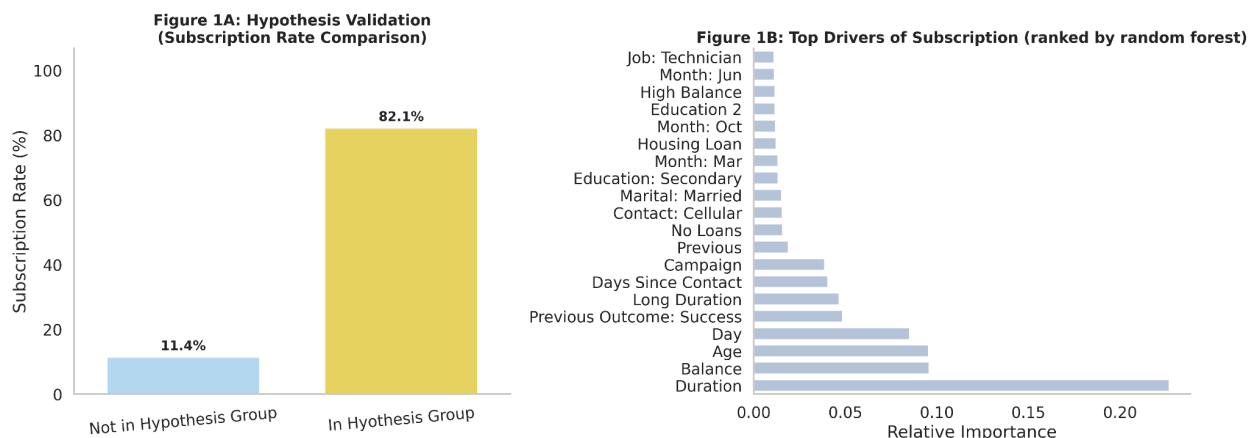
The UCI Bank Marketing dataset contains information from over 40,000 telephone contacts made during several marketing campaigns. Each row represents an individual client and includes demographic characteristics (such as age, marital status, and education), financial indicators (including account balance, housing loan, and personal loan), and details about the most recent contact (such as contact type, month, and duration). The target variable indicates whether a client subscribed to a term deposit.

Before analysis, the dataset was cleaned to remove obvious inconsistencies and to encode categorical features in a format suitable for machine learning. Several feature-engineering steps were conducted to better capture behavioral patterns. For example, contact duration, although highly predictive, can only be known after a call has ended; therefore, we considered its importance but interpreted it cautiously as a post-contact feature. Variables with many missing values or minimal variance, especially those that did not meaningfully distinguish between subscribers and non-subscribers, were omitted to avoid noise. Additionally, we used one-hot encoding to generate binary variables for education levels, employment groups, and communication channels, enabling the model to isolate and interpret each category's specific effect.

A limitation of the dataset is that it does not include psychographic or attitudinal information, which real-world marketers often rely on. Similarly, income is not directly observed, so our model approximates it using related variables such as job type and account balance. These

constraints mean that some potentially influential factors are absent from the analysis. Also, the severe class imbalance—only about 12% subscribed to a long-term deposit—makes the classifier prone to favoring the majority class. To address this, we applied SMOTE, which creates synthetic minority examples by interpolating between similar “Yes” cases, helping the model learn a more balanced decision boundary.

To explore the factors driving subscriptions, we analyzed conversion rates across client segments. Figure 1A illustrates that clients in the hypothesis group—those with high balance, no loans, longer past engagement, and previous success—convert far more often than the rest (82.1% vs. 11.4%). This indicates that specific client traits align with higher subscription likelihood. Figure 1B shows why this difference appears: several behavioral and contextual variables—not just client profile—contribute a meaningful predictive signal. Factors such as call duration, recency of engagement, and prior campaign outcomes all show substantial association with subscription. This suggests that we should prioritize clients with strong prior engagement and reach them when the context is right—specifically, by optimizing call timing and sequence (utilizing factors like “Day” and “Month” importance) through effective channels (i.e cellular).



With the knowledge that certain client features predict subscription better than others, we trained machine learning models to operationalize our targeting strategy. As shown in Figure 2A, the XGBoost model casts a wide net, finding most potential subscribers but with low precision—meaning over half the calls it generates are wasted. The Random Forest model is more precise; it finds slightly fewer total subscribers, but over half of the people it flags actually subscribe, making it far more efficient.

Figure 2B provides a tangible example by simulating the operational outcome based on the model score, assuming a fixed budget of 1,000 calls. The difference is stark: the random baseline

yields only 116 subscriptions, while the models increase sales by over five times, achieving 604 (XGBoost) or 583 (Random Forest) subscriptions.

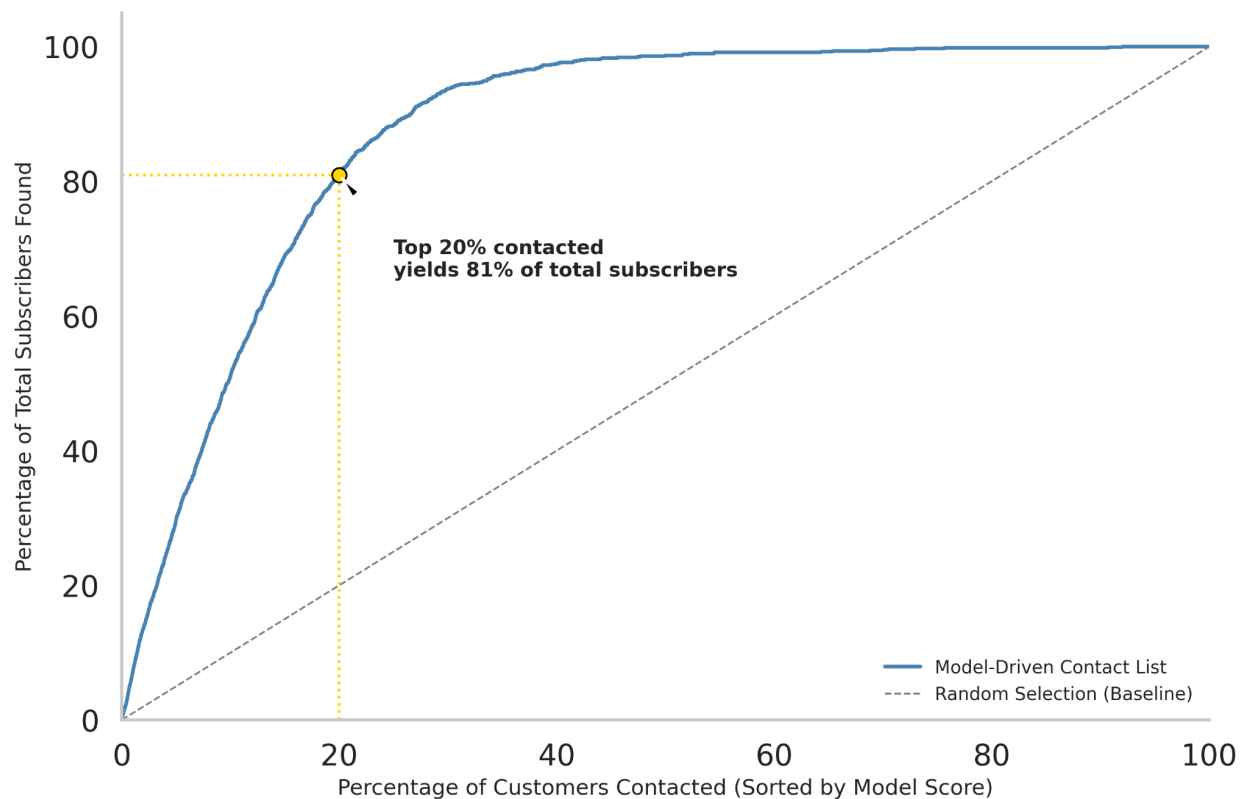
While the choice of the best model can depend on whether the campaign's priority is to maximize conversions and avoid missing potential subscribers or to reduce the cost of unnecessary calls, the balance between the two is important. Precision is paramount for the campaign because marketing resources are finite. Every call to an uninterested client wastes time, money, and risks customer irritation. The Random Forest model provides the optimal balance, concentrating our team's effort on the most promising clients and ensuring the highest return on every call made.



What do we recommend?

Based on the high ROC-AUC scores established in Figure 2, which confirm the model's excellent ranking ability, we recommend implementing a **"Skimming Strategy"** with a precise cut-off point. Figure 3 below demonstrates that targeting the **Top 20%** of the model-scored list captures **81%** of the total available subscribers. This is the most efficient and practical point, as continuing to contact the remaining 80% of the list provides diminishing returns for significant increases in operational cost. This strategy allows the bank to capture maximum revenue while drastically reducing contact expenditure compared to the Random Selection Baseline.

Figure 3: The "Skimming" Strategy (Cumulative Gain Curve)



Discussion

The findings of this study reinforce and extend existing research on bank direct marketing by showing that customer engagement and interaction history are substantially more informative than static demographic characteristics when predicting term-deposit subscription. Prior studies using the Bank Marketing dataset have similarly identified call duration and previous campaign outcomes as dominant predictors (Moro, Cortez & Rita, 2014; Moro et al., 2011). Our results are consistent with these conclusions, while further demonstrating that their predictive relevance persists across different modeling approaches and becomes more pronounced when interaction effects are explicitly considered.

An important interpretive contribution of this work lies in how engagement-based variables are contextualized. Previous literature often highlights call duration as a strong predictor but cautions against causal interpretation, since duration is observed after contact begins (Moro et al., 2011). Rather than treating duration as a standalone driver, our analysis frames it as a proxy for conversational engagement whose effect depends on prior responsiveness and contact context. This interaction-based view helps reconcile earlier findings by explaining why duration

performs well in predictive models without implying that simply prolonging calls would increase subscriptions.

The model comparison highlights a practical dimension that is underemphasized in much of the academic literature: the operational trade-off between recall and precision under limited resources. Models optimized for recall may identify more potential subscribers but at the cost of substantial wasted outreach. In contrast, models with higher precision better align with real-world telemarketing constraints, where minimizing unnecessary calls is critical. This finding echoes recent applied marketing research advocating cost-aware evaluation over purely statistical performance measures.

The cumulative-gain analysis situates these results within the broader response-modeling literature, which has long observed that customer responsiveness tends to be highly concentrated in a small segment of the population. Our findings empirically support this pattern, revealing a clear Pareto-like structure in subscription likelihood. From a managerial perspective, this validates ranked targeting strategies and provides a straightforward operational rule: focusing on a narrow, high-scoring subset of clients yields most of the attainable value, while broader outreach produces diminishing returns.

Beyond predictive performance, this study illustrates how machine-learning models complement heuristic segmentation rather than replacing it. While rule-based indicators of financial capacity and loan status identify a highly responsive group, they fail to capture customers whose behavioral patterns signal readiness despite weaker financial markers. By integrating engagement, timing, and interaction history, predictive models uncover these nuanced cases and translate descriptive insights from earlier milestones into scalable decision rules.

Several limitations remain. The analysis relies on observational data from a specific temporal and geographic context, limiting causal inference and generalizability. Unobserved factors such as agent effectiveness or customer sentiment may also influence outcomes. Nonetheless, the alignment of our findings with established literature, combined with their operational framing, suggests that the conclusions are robust within the scope of the study.

Overall, this work demonstrates that combining behavioral insight with interaction-aware predictive modeling enables more efficient and strategically grounded direct marketing. By aligning model outputs with operational constraints, the study offers a practical framework for deploying data-driven targeting in real-world banking contexts.

References

Moro, S., Laureano, R., & Cortez, P. (2011). *Using data mining for bank direct marketing: An application of the CRISP-DM methodology*.

Moro, S., Cortez, P., & Rita, P. (2014). A data-driven approach to predict the success of bank telemarketing. *Decision Support Systems*, 62, 22–31.